



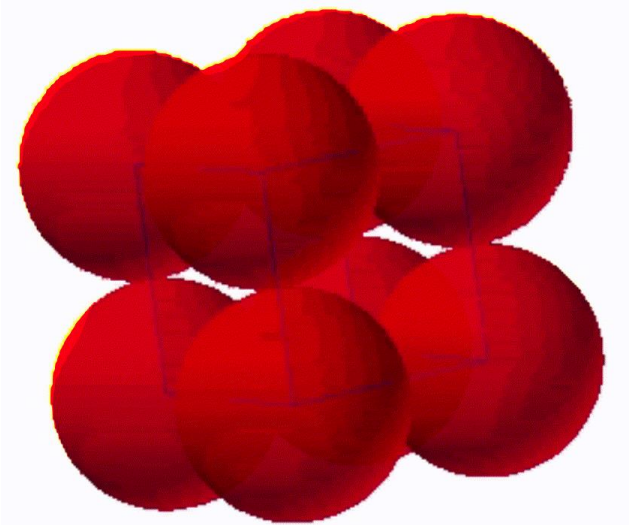
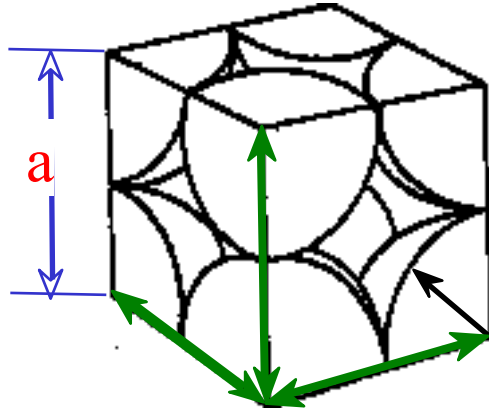
Outline:

- Review of Crystal structure
- Stacking Sequence
- Theoretical Density
- Crystallographic Directions and Planes
 - *examples*



Review: *Crystal Structure*

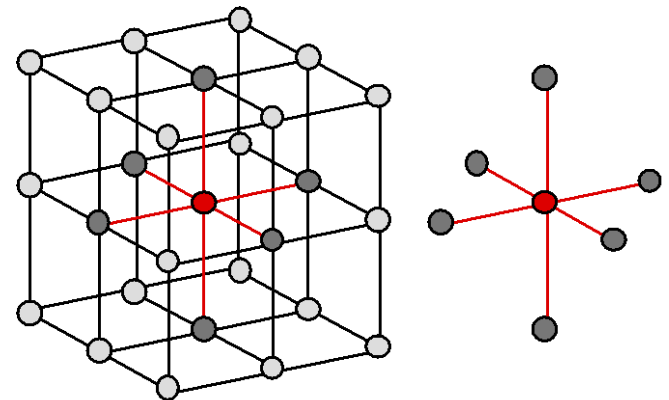
- Close-packed directions are



This is a unit cell

- Contains =
- Coordination # = (# nearest neighbors)

$$\text{APF} = \frac{1 \frac{4}{3} \pi (0.5a)^3}{a^3}$$



- APF for a simple cubic structure =

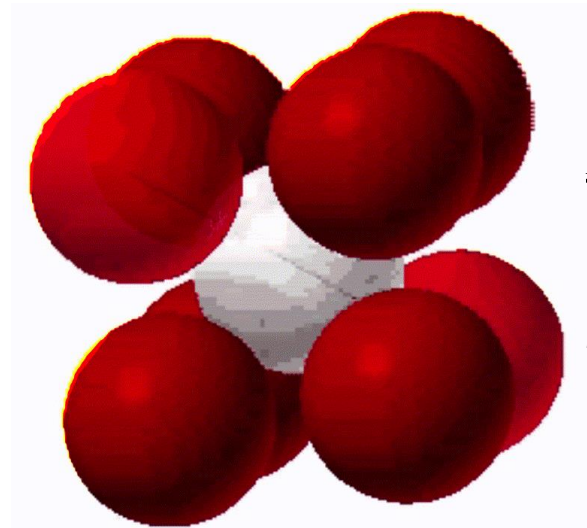


Review: Crystal Structure

- Close packed directions are
- Coordination # =

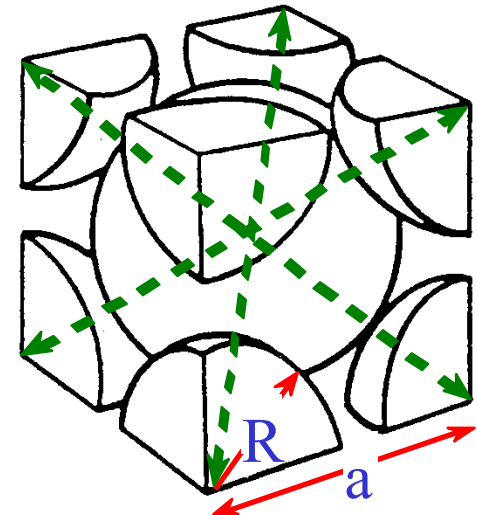
$$\text{APF} = \frac{2 \cdot \frac{4}{3} \pi \left(\frac{\sqrt{3}a}{4} \right)^3}{a^3}$$

- APF for a BCC structure =



Note: All atoms are identical; the center atom is shaded differently only for ease of viewing.

This is a unit cell

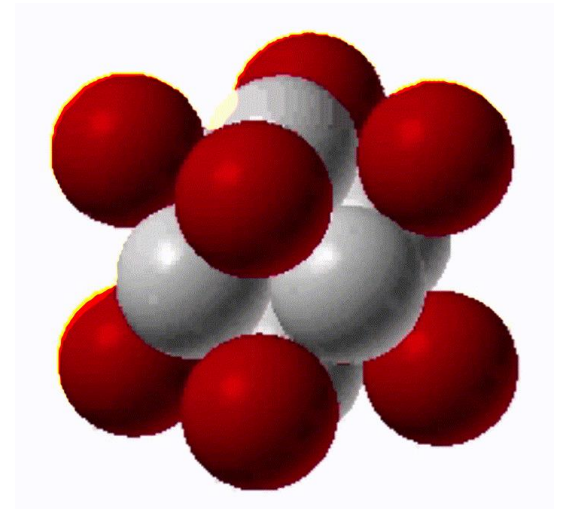




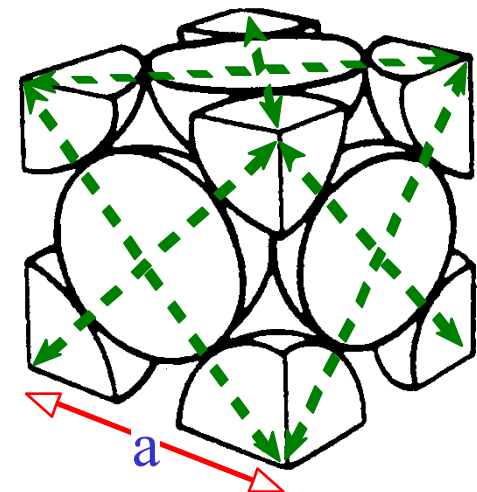
Review: *Crystal Structure*

- Close packed directions are
- Coordination # =

$$\text{APF} = \frac{4 \cdot \frac{4}{3} \pi \left(\frac{\sqrt{2}a}{4} \right)^3}{a^3}$$



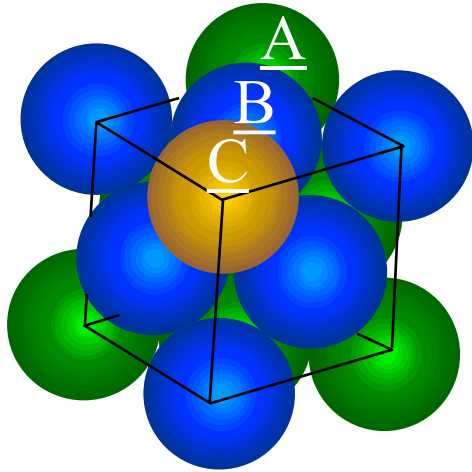
This is a unit cell



- APF for a FCC structure =



FCC Stacking Sequence

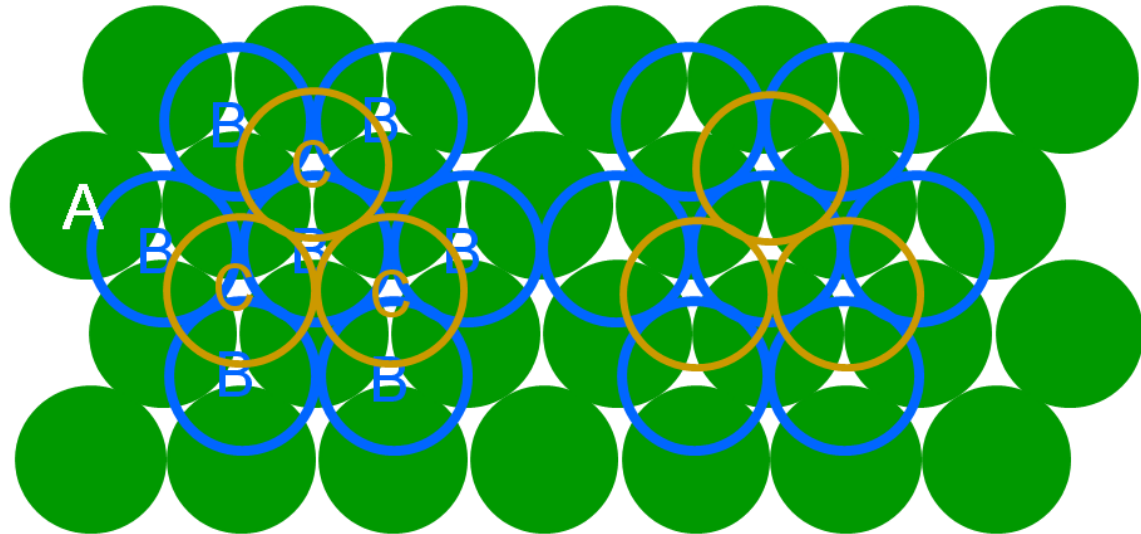


- FCC Unit Cell

A sites

B sites

C sites



2D Projection

- ABCABC...etc Stacking Sequence



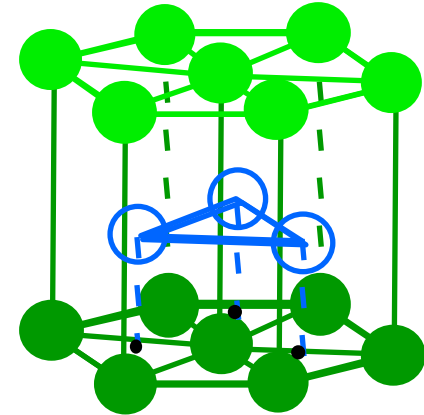
Review: *Crystal Structure*

- Coordination # =
- APF =

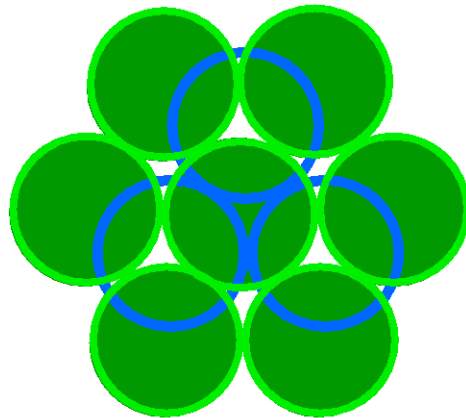
A sites

B sites

A sites



This is a unit cell

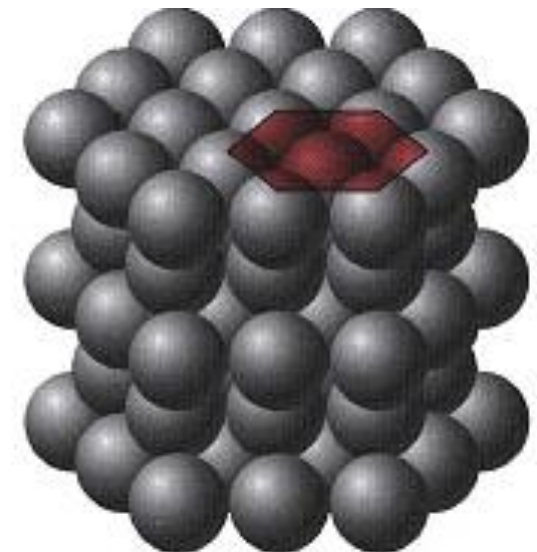


Top layer

Middle layer

Bottom layer

2D Projection



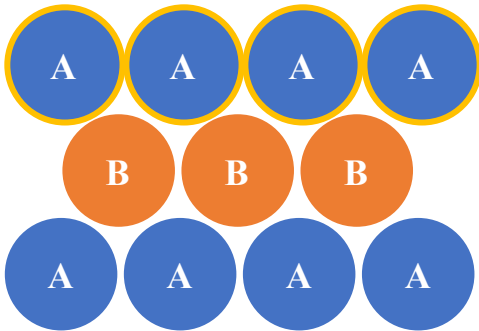
- Stacking Sequence:



HCP vs. FCC: *the Third Layer Decides*

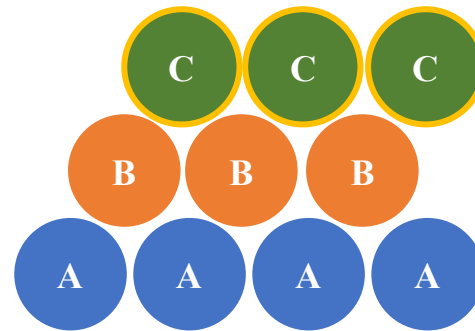
Both start the same way: a close-packed A layer, then a B layer in the hollows.

HCP — ABAB



Third layer sits directly above the A atoms. The C hollows stay empty, so the pattern repeats every two layers: ABAB...

FCC — ABCABC



Third layer shifts into the empty C hollows — a new position. Only the fourth layer returns to A: ABCABC...

Same packing, different order: $CN = 12$ and $APF = 0.74$ for both. The stacking sequence is what differs — and it is why FCC and HCP deform so differently.



Example Problem

If you know

- the crystal structure,
- the atomic radius
- the atomic weight,

you can calculate the density of a particular material

Example:

Copper has an atomic radius 0.128 nm an FCC crystal structure and an atomic weight of 63.5 g/mol. Calculate its density.



Crystallographic Directions, and Planes

Now that we know how atoms arrange themselves to form crystals, *we need a way to identify directions and planes of atoms.*

Why?

- ✓ Deformation under loading (.....) occurs on certain crystalline planes and in certain crystallographic directions.
- ✓ Before we can predict how materials fail, we need to know what modes of failure are more likely to occur.
- ✓ Other properties of materials (*electrical conductivity, thermal conductivity, elastic modulus*) can vary in a crystal with orientation.

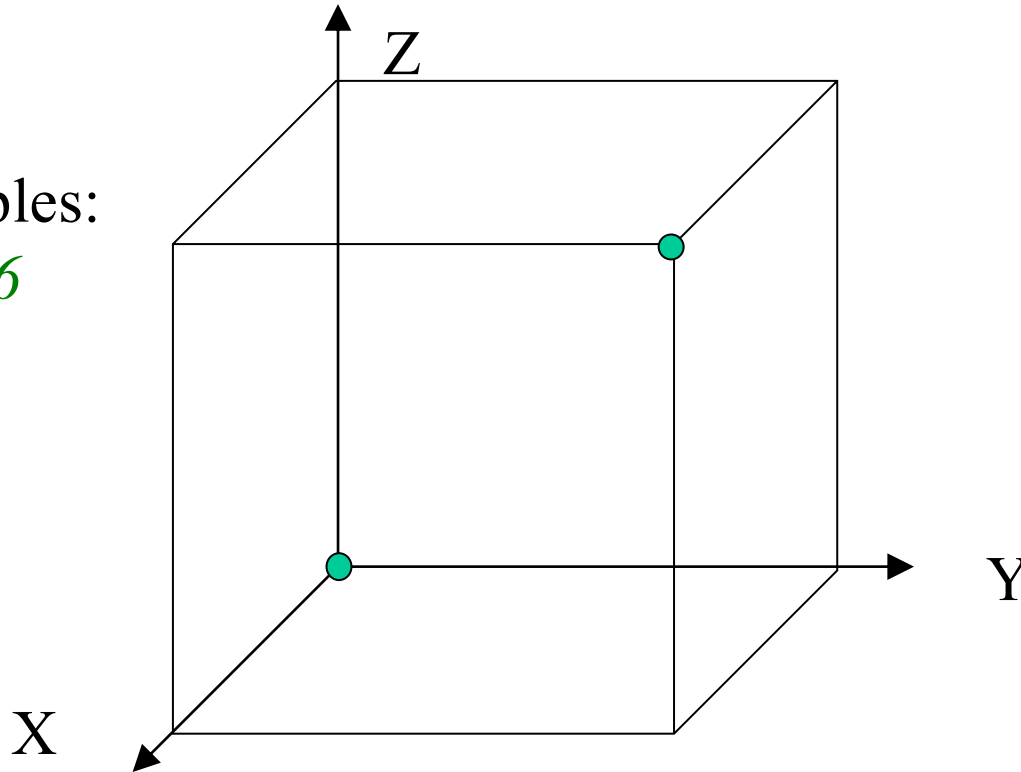
Directions and planes are described using three integers -
..... Indices



Point Coordinates

Point position specified in terms of its coordinates as fractional multiples of the unit cell edge lengths

See examples:
3.5 and 3.6





General Rules for Lattice Directions, Planes & Miller Indices

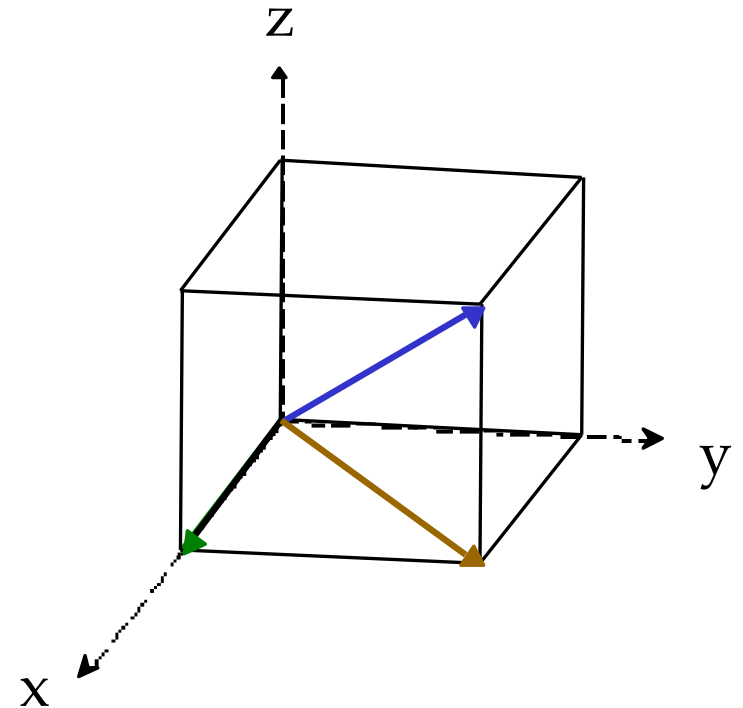
- Miller indices used to express lattice planes and directions
 - x, y, z are the axes (on arbitrarily positioned origin)
 - *in some crystal systems these are not mutually $\perp$*
 - a, b, c are lattice parameters (*length of unit cell along a, b, c sides*)
 - h, k, l are the Miller indices for planes and directions - expressed as planes: (hkl) and directions: $[hkl]$
 - Conventions for naming
 - There are **NO COMMAS** between numbers
 - Negative values are expressed with a bar over the number
 - *Example: -2 is expressed $\bar{2}$*
- Crystallographic direction:
 - $[123]$
 - $[100]$
 - ... etc.



Miller Indices for Directions

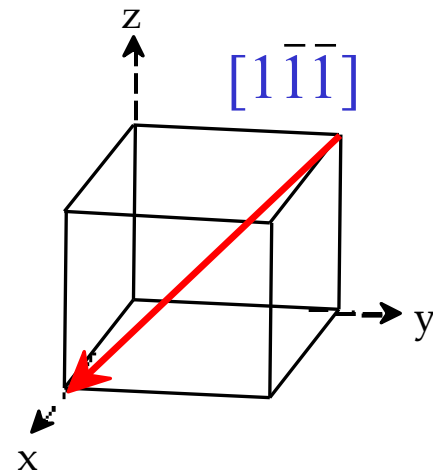
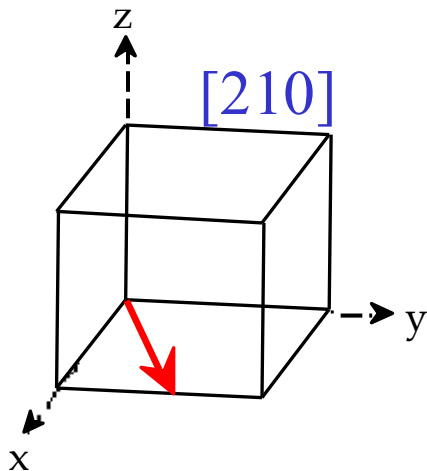
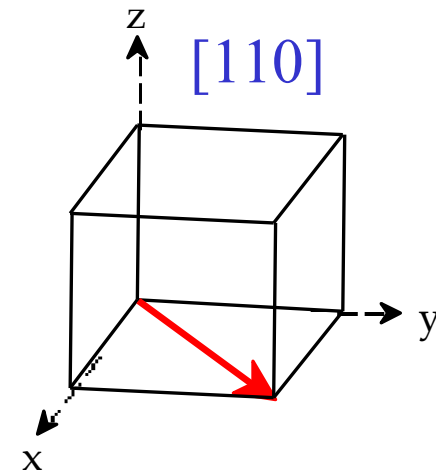
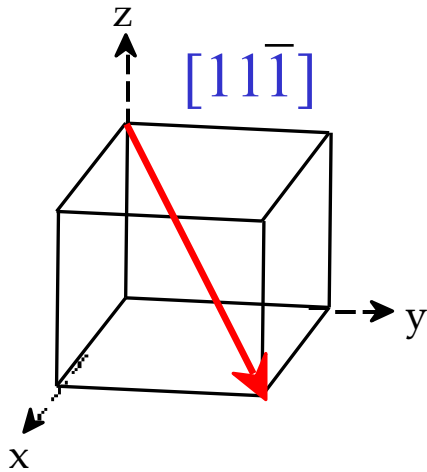
Method

- Draw vector, define tail as origin.
- Determine length of the vector **projection** in unit cell dimensions, a , b , and c
- Remove fractions by multiplying by smallest possible factor
- Enclose in square brackets



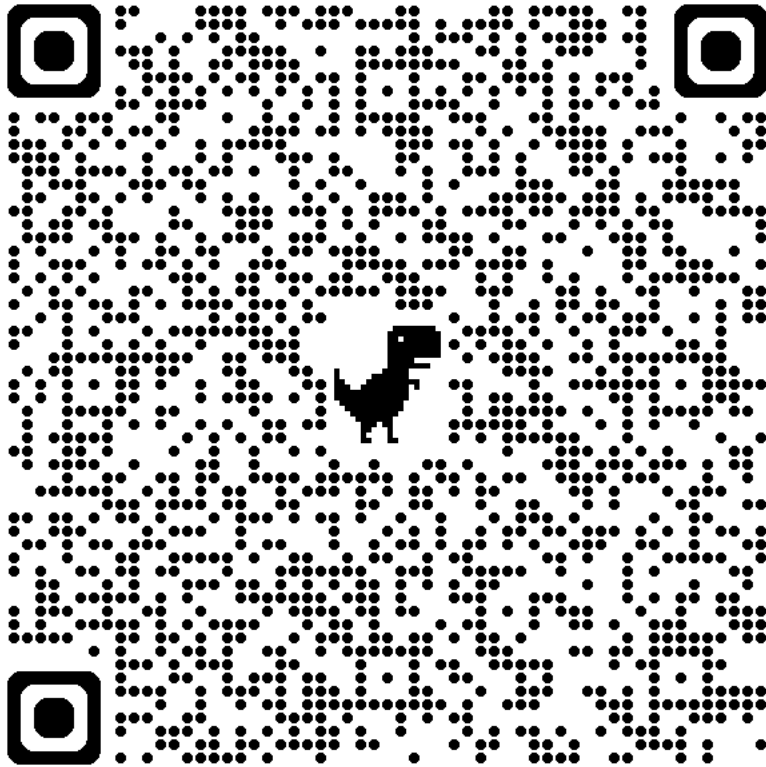


Example - Naming Directions



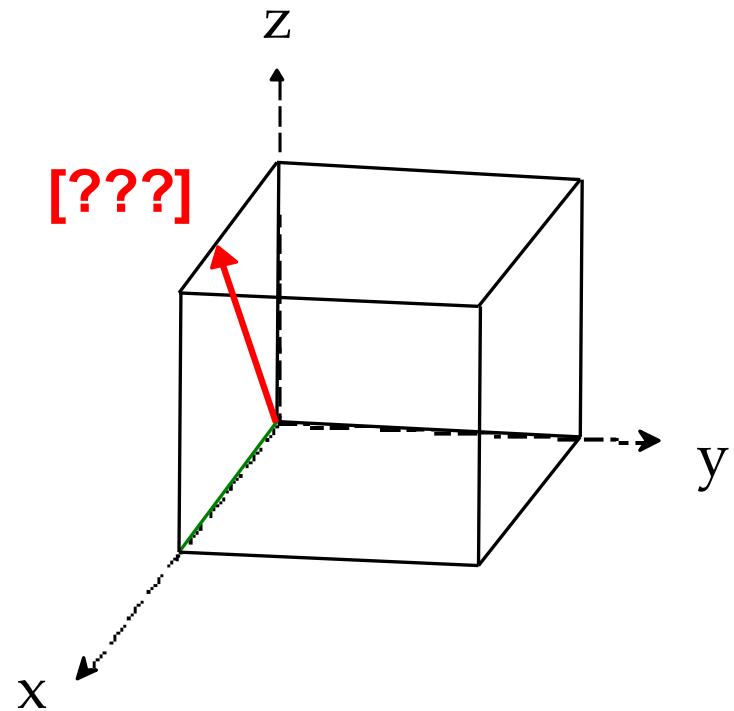


Miller Indices for Directions



Use the QR code shown in the classroom. This one won't work.

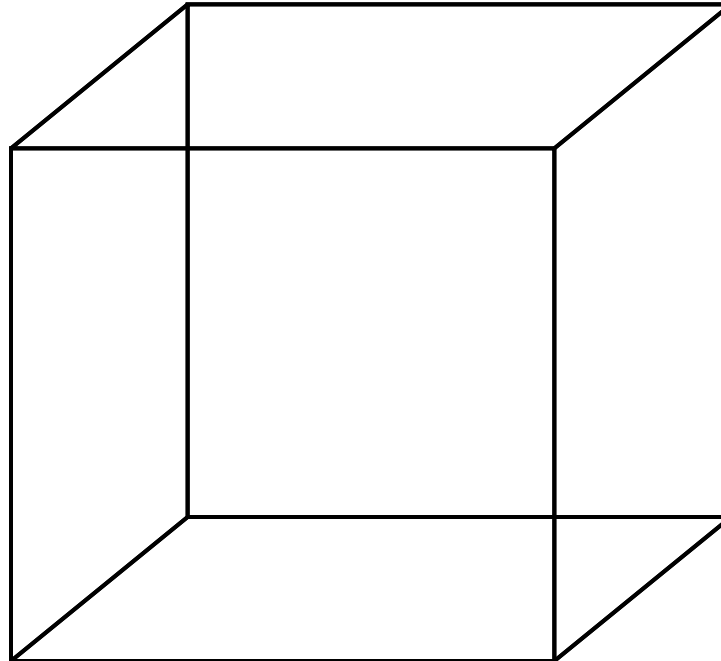
What is [? ? ?]





Example - Drawing Directions

- Draw $[11\bar{2}]$ and $[\bar{1}\bar{1}\bar{1}]$

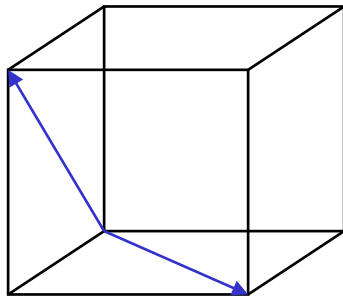


The HW problems will give you more practice with directions



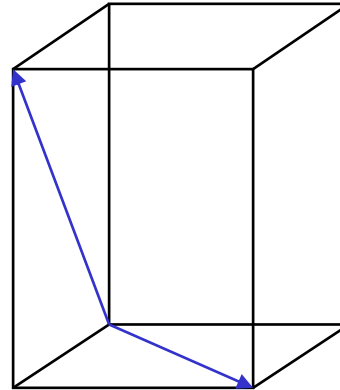
Families of Directions

- Equivalence of directions



$$[101] = [110]$$

cubic



$$[101] \neq [110]$$

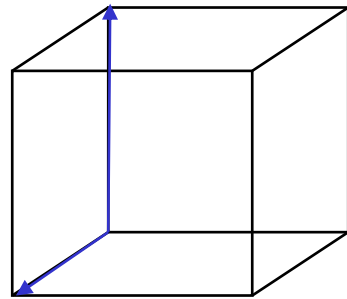
tetragonal

- $\langle 123 \rangle$ Family of directions
 - $[123], [213], [312], [132], [231], [321]$
– *only in a cubic crystal*

*In the **cubic system** directions having the same indices regardless of order or sign are *equivalent*.*



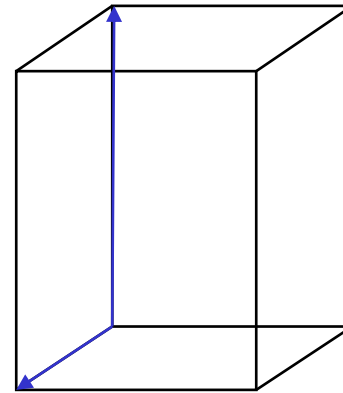
Isotropy vs. Anisotropy in Single Crystals



$$[100] = [001]$$

cubic

Isotropic



$$[100] \neq [001]$$

tetragonal

Anisotropic

- Properties are independent of direction → isotropic material
- Directionality of properties → anisotropic material



Miller Indices for Planes

- (hkl) Crystallographic plane
- $\{hkl\}$ Family of crystallographic planes
 - e.g. (hkl) , (lhk) , (hkl) ... etc.

In the cubic system planes having the same indices regardless of order or sign are equivalent

- **Hexagonal** crystals can be expressed in a four index system $(u\ v\ t\ w)$
 - *Can be converted to a three index system using formulas*

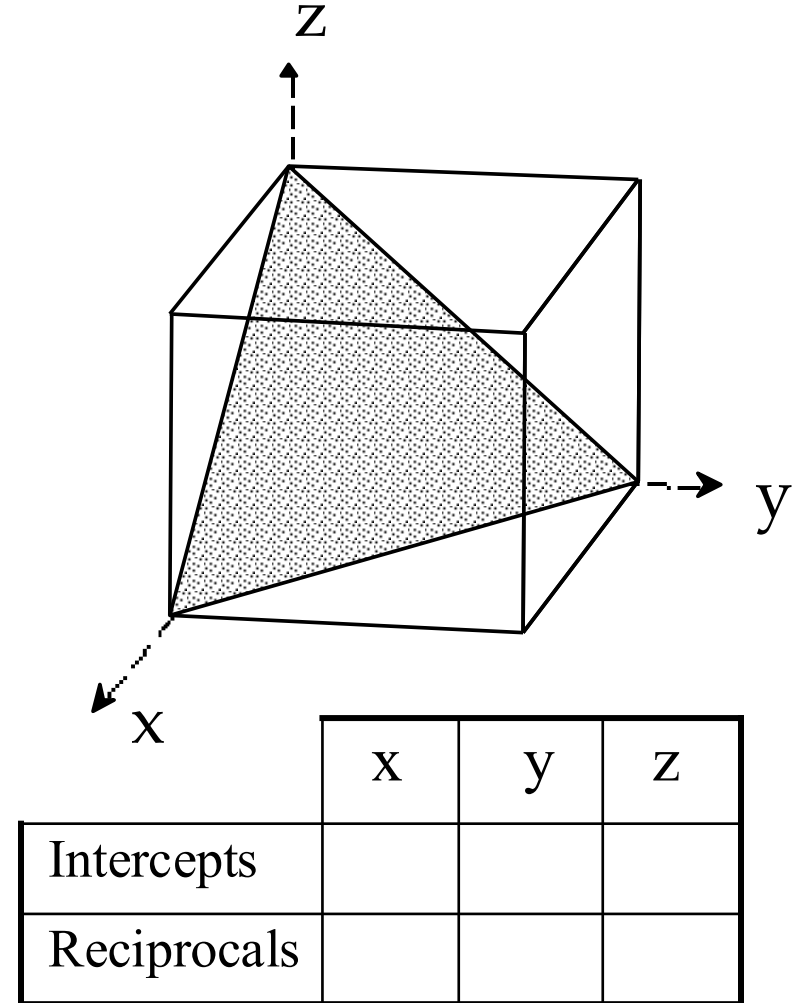


Miller Indices for *PLANES*

Method

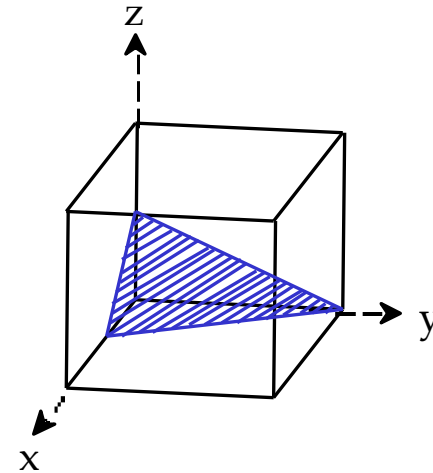
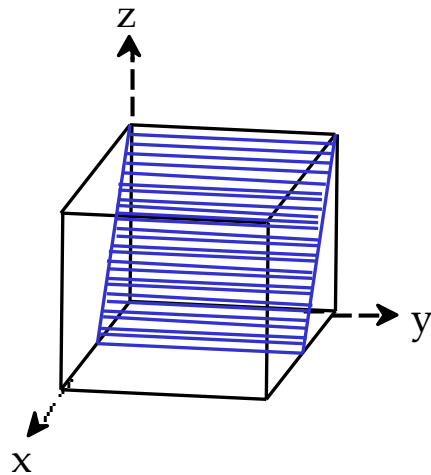
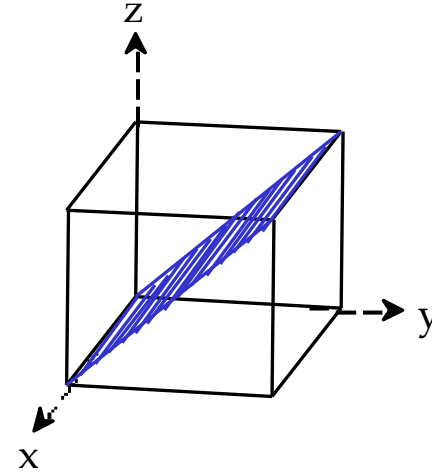
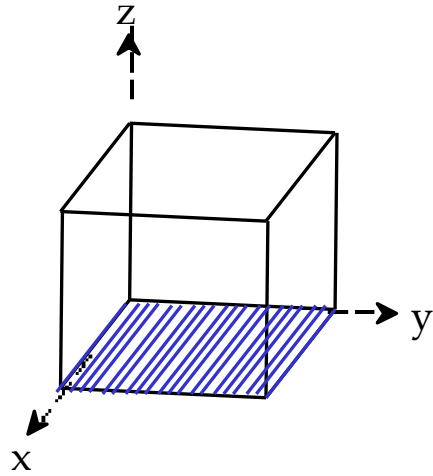
- If the plane passes through the origin, select an equivalent plane or move the origin
- Determine the intersection of the plane with the axes in terms of a, b, and c
- Take the reciprocal ($1/\infty = 0$)
- Convert to smallest integers (*optional*)
- Enclose by parentheses

see example 3.10





Crystallographic Planes





Next time:

Linear and Planner Densities